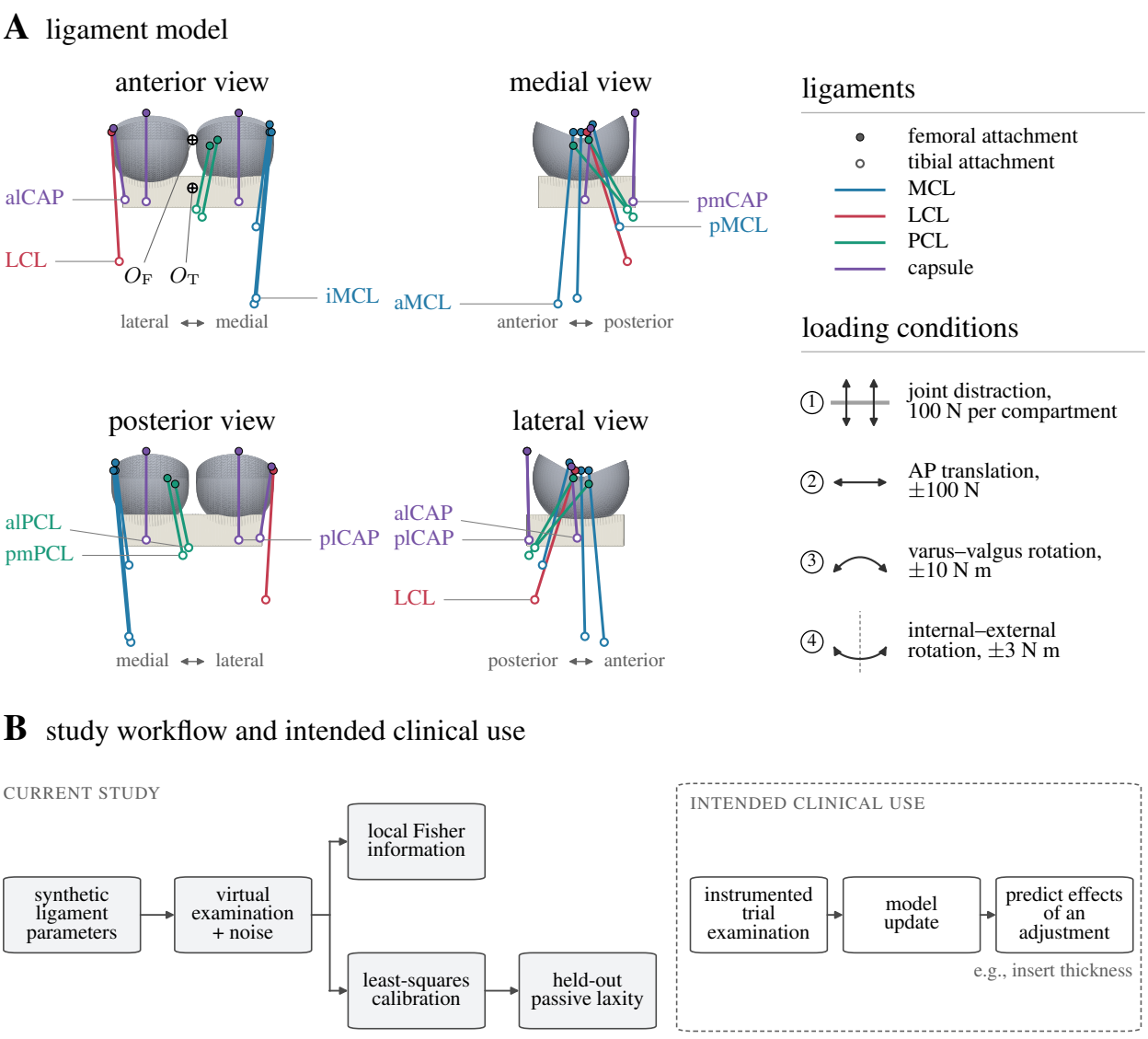

A ligament model
anterior view
alCAP
LCL
O_F
O_T
iMCL
lateral ↔ medial
medial view
pmCAP
pMCL
aMCL
anterior ↔ posterior
posterior view
alPCL
pmPCL
plCAP
medial ↔ lateral
lateral view
alCAP
plCAP
LCL
posterior ↔ anterior
ligaments
femoral attachment
tibial attachment
MCL
LCL
PCL
capsule
loading conditions
① joint distraction, 100 N per compartment
② AP translation, ±100 N
③ varus–valgus rotation, ±10 N m
④ internal–external rotation, ±3 N m
B study workflow and intended clinical use
CURRENT STUDY
synthetic ligament parameters
virtual examination + noise
local Fisher information
least-squares calibration
held-out passive laxity
INTENDED CLINICAL USE
instrumented trial examination
model update
predict effects of an adjustment
e.g., insert thickness